

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

Our goal is to measure photometric variability in ultracool brown dwarfs and determine whether their brightness changes over time. Brown dwarf variability can provide information about their atmospheres and changing cloud features. We plan to observe 2MASSW J0036159+182110 and 2MASSW J0149090+295613 with the 90-inch Bok Telescope on Kitt Peak using the 90 Prime instrument. We will take repeated 200 s exposures for approximately 8 hours to produce time-series light curves. Previous observations of brown dwarfs, such as Heinz et al. (2014), used **200 s exposures** with the F814W filter and achieved approximately **1.2% RMS photometric precision** for a target with similar brightness, supporting 200 s as a reasonable choice for our observations. The long observing period should allow us to track brightness changes over many consecutive measurements rather than relying on individual observations.

The resulting images will be calibrated and analyzed using differential photometry, comparing the brightness of the brown dwarfs with nearby comparison stars. This will help reduce the effects of changing observing conditions and improve the reliability of the measured brightness variations. We will examine the resulting light curves for measurable changes in brightness. This will allow us to determine whether the targets show evidence of photometric variability. These observations will help characterize variability in ultracool brown dwarfs and provide information about changes in their atmospheres.

Telescope Requested (0.5 points): 90-inch Bok Telescope

Instrument (0.5 points): 90 Prime

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations? We prefer to keep the target at least 50° away from the Moon. This will reduce the effect of moonlight on the sky background and improve the sensitivity of our photometry.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

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Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
2MASSW J0036159+182110	9.0671	18.3529	17.1922 19.5066	L3.5	~8.67h
2MASSW J0149090+295613	27.2875	29.9368	17.7745 19.9889	M9.5	~8h

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

We request approximately 8 hours of observing time with 200 s exposures. The exposure time should provide sufficient signal-to-noise for our relatively faint targets while allowing us to obtain many measurements throughout the observing period. Previous brown dwarf observations have also used 200 s exposures, providing a reference for our observing strategy.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

These observations will provide repeated measurements of the brown dwarfs' brightness over several hours. By analyzing the resulting light curves with differential photometry, we can identify changes in brightness while accounting for changes in observing conditions. Detecting significant variations will allow us to determine whether the targets are photometrically variable and better understand their atmospheric behavior.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If weather prevents us from obtaining sufficient observations, we can use archival observations of brown dwarfs to perform time-series photometry. We could analyze existing images to search for brightness variations and compare the results with our expected variability.